4.7 Diagnostics Button

This button will bring up a Diagnostics Display Window.This window is used to study the geophysical performance of acquisition geometries in detail. The window uses either CMP or DMO hit structures, which are generated in the binning menu

The area which has been binned is displayed in the upper left corner of the parameters window. Any sub-area may be selected with the "Plot Min/Max Local Bin/Trk" parameters or by using the left mouse button to draw a rubberband box (with the Button Mode set to Bin Area). This allows the user to "zoom in" on a given area.

Color scale selection The user controls the number of colors (up to 20), and has the option of controlling the color scaling. The color scale may be edited with the colormap editor. The number of colors is selected in the parameters menu.

Pixmap size The user can change the size of the diagnostics pixmap by changing the "Picture Width/Height" parameters and then clicking "Clear/Resize". Optionally the user can use the left mouse button to draw a rubberband box to delimit the area to be zoomed

Global offset limitations Only traces between the minimum and maximum offsets specified are used in the diagnostic calculations. In addition, traces must be less than the stretch-factor limited offset:

$$X_{\max} = t_0 V \sqrt{s^2 - 1} \qquad \text{(EQ 7)}$$

These parameters are entered at the bottom of the diagnostics main menu. If stretch-factor limiting is not desired, set the zero-offset time to 100, the velocity 100000, and the stretch factor to 10.

4.7.1 Offset diagnostics

Diagnostics which have additional parameters are indicated with a "-p". These parameters are accessed in the "Parameters..." popup menu.

Multiplicity, Minimum Offset, Maximum Offset

Usage
- Set the offset limitations in the main menu.
- Click on "Multiplicity", "Minimum Offset", or "Maximum Offset".

NOTE: To ensure that proper binning has been accomplished the multiplicity diagnostic should be checked prior to performing other diagnostics.

Offset Uniformity.The number of intervals, of size "Interval distance", between the maximum and minimum offsets which contain at least one trace. This measure depends on the somewhat arbitrary position of the interval boundaries, and can either undercount or overcount the actual offset uniformity.

Usage
- Set the offset limitations in the main menu.
- Set the "interval distance" parameter (Parameters menu).
- Click on "Offset Uniformity".

Discrimination Multiplicity. The traces in each cell are sorted according to offset. Starting with the trace with the minimum offset, traces are only counted if the offset difference with the previous trace is greater than the "Discrimination distance". If trace N is not counted for this reason, then trace N+1 is only counted if the offset difference with trace N-1 is greater than the "Discrimination distance", etc. This diagnostic tends to yield a measure which is too pessimistic.

Usage

• Set the offset limitations in the main menu.
• Set the "discrimination distance" parameter (Parameters menu).
• Click on "Discrimination Multiplicity".

Effective Fold. This diagnostic was designed to overcome the flaws of the offset uniformity and discrimination multiplicity diagnostics.The effective fold is found by first calculating the fraction of the offset range between (Min offset - δr) and (Max offset + δr) that is within δr of a trace, where

$$\delta r = \frac{\text{AverageSeparation}}{2} \qquad \text{(EQ 8)}$$

This fraction is then multiplied by the "ideal" multiplicity:

$$\text{EffectiveFold} = \left(\frac{(\text{MaxOffset} - \text{MinOffset})}{\text{AverageSeparation}} + 1 \right) \times \text{fraction} \qquad \text{(EQ 9)}$$

Usage
• Set the offset limitations in the main menu.
• Set the "average separation" parameter (Parameters menu).
• Click on "Effective fold".

Delta-offset RMS.

Let x_i be the offset of trace i, with $x_i \geq x_{i-1}$, and N be the number of traces in the cell between the maximum and minimum offsets. Then $\langle \delta x \rangle_{\text{RMS}}$ is given by

$$\langle \delta x \rangle_{\text{RMS}} = \sqrt{\frac{1}{N-1} \sum_{n=2}^{N} (x_n - x_{n-1} - \langle \delta x \rangle)^2} \qquad \text{(EQ 10)}$$

$$\text{where} \qquad \langle \delta x \rangle = \frac{x_N - x_1}{N-1}$$

Usage
• Set the offset limitations in the main menu.
• Click on "Delta-offset RMS".

Unique Receiver Lines.

This diagnostic is *only* useful for surveys which have receiver lines parallel to the x-axis and have not been perturbed by culture/terrain. It is intended to give some idea the effect moving receiver lines will have on the survey. For example, if a survey design has only two lines per cell, then moving one receiver line in the crossline direction could halve the fold in some cells. This diagnostic is also a simple measure of receiver line statics coupling, which should improve as the number of receiver lines per cell increases.

Usage
• Set the offset limitations in the main menu.
• Click on "Unique Receiver Lines".

<Inline Offset> Detects cell-to-cell variations in the average inline component of the offet.

$$\langle \text{InlineOffset} \rangle = \frac{1}{N} \sum_{n=1}^{N} (r_x - s_x) \qquad \text{(EQ 11)}$$

where r_x is the inline coordinate of the receiver and s_x is the inline coordinate of the source.

Usage
- Enter the offset limitations (main menu)
- Click on "<Inline Offset>".

<Crossline Offset>

Detects cell-to-cell variations in the crossline component of the offset. This can be useful for detecting cable feather in marine surveys.

$$\langle \text{CrosslineOffset} \rangle = \frac{1}{N} \sum_{n=1}^{N} (r_y - s_y) \qquad \text{(EQ 12)}$$

where r_y is the crossline coordinate of the receiver and s_y is the crossline coordinate of the source.

Usage
- Enter the offset limitations (main menu)
- Click on "<Crossline Offset>".

Offset Range.

Detects cells with a large minimum offset and a small maximum offset.

Range = maximum offset - minimum offset (EQ 13)

Usage
- Enter the offset limitations (main menu)
- Click on "Offset range".

Largest Gap.

This diagnostic finds the largest "hole" in the range of offsets between all adjacent valid traces.

Usage
- Enter the offset limitations (main menu)
- Click on "Largest Gap".

4.7.2 Coherent noise suppression diagnostics

All of the following diagnostics are intended to give some idea how much coherent noise will be attenuated by the processing sequence NMO correction => stack. They do not predict suppression if the processing sequence includes DMO.

Multiple Suppression by Stacking.

This diagnostic is discussed in references[1,2,3], The formula used to calculate it is

$$R = \frac{1}{f_{\max} - f_{\min}} \int_{f_{\min}}^{f_{\max}} df |R(f)|^2 \qquad \text{(EQ 14)}$$

where

$$\Re(f) = \frac{1}{N}\sum_{n=1}^{N} e^{2\pi i f t_n} \qquad \text{and} \qquad t_n = \sqrt{t_0^2 + \left(\frac{x_n^2}{V_M^2} - \frac{x_n^2}{V_P^2}\right)} \qquad \text{(EQ 15)}$$

Usage

- Enter the offset limitations (main menu).
- Enter the multiple velocity (parameters menu)
- Enter the minimum and maximum frequencies (parameters menu)
- Set "Include Arrays?" to "<> NO" (parameters menu).
- Enter the primary velocity, zero-offset time, and the stretch factor (main menu)
- Click on "Multiple Supp.".

Multiple suppression by stacking (with source and receiver array effects). The apparent velocity of the multiple at the surface is a function of the offset:

$$V_A = V_M\sqrt{1 + \frac{t_0^2 V_M^2}{X^2}} \qquad \text{(EQ 16)}$$

Therefore, the effective wavenumber at the source and receiver arrays is a function of the frequency and offset:

$$k(X,f) = \frac{2\pi f X}{V_M\sqrt{X^2 + t_0^2 V_M^2}} \qquad \text{(EQ 17)}$$

The suppression can be calculated by integrating the following equation numerically:

$$R(f) = \frac{1}{N}\sum_{n=1}^{N} e^{2\pi i f t_n} R_R(k,\theta_n) R_S(k,\theta_n)$$

$$R = \frac{1}{f_{\max} - f_{\min}} \int_{f_{\min}}^{f_{\max}} df |R(f)|^2 \qquad \text{(EQ 18)}$$

where $R_R(k,\theta)$ is the response of the receiver array and $R_S(k,\theta)$ is the response of the source array. They are described in chapter 8. This calculation uses the far-field approximation.

Usage

- Enter the offset limitations (main menu)
- Enter the multiple velocity (parameters menu)
- Enter the minimum and maximum frequencies (parameters menu)
- Set "Include Arrays?" to "<> Yes" (parameters menu).
- Enter the primary velocity, zero-offset time, and the stretch factor (main menu)
- Make sure that the source and receiver arrays are specified in the array submenus in the Array window.
- Click on "Multiple Supp".

Stack response. This diagnostic predicts average signal and noise suppression over a range of wavelengths if the traces are stacked together with no NMO correction. This approximation follows the so-called "stack-array" approach.

$$R = \frac{1}{k_{max} - k_{min}} \int_{k_{min}}^{k_{max}} dk \left| \frac{1}{N} \sum_{n=1}^{N_{noise}} e^{ikx_n} \right|^2 \qquad \text{(EQ 19)}$$

where

$$k_{max} = \frac{2\pi}{\lambda_{min}} \text{ and } k_{min} = \frac{2\pi}{\lambda_{max}} \qquad \text{(EQ 20)}$$

Usage

- Enter the offset limitations (main menu).
- Enter the minimum and maximum wavelengths (parameters menu)
- Set "Include Arrays?" to "<> No" (parameters menu).
- Enter the minimum and maximum offset ranges within which the noise with the above wavelength appears (parameters menu).
- Click on "Stack Response".

The minimum and maximum offsets over which the noise appears provides a method of analyzing the response to offset-limited noise. For example, if the surface wave noise cone extends from 1000m to 1750m at the objective time, data recorded outside that offset range will be noise-free. The N_{noise} in the above equation represents the number of traces within the noise offset range, whereas N represents the number of traces actually stacked given the offset limitations in the main menu.

Stack response with array effects. This diagnostic incorporates the source and receiver arrays into the stack response using the far field approximation

$$R(k) = \frac{1}{N} \sum_{n=1}^{N_{noise}} e^{ikx_n} R_R(k, \theta_n) R_S(k, \theta_n) \qquad \text{(EQ 21)}$$

The response is found by integrating the above numerically:

$$R = \frac{1}{k_{max} - k_{min}} \int_{k_{min}}^{k_{max}} dk |R(k)|^2 \qquad \text{(EQ 22)}$$

Usage

- Enter the offset limitations (main menu).
- Enter the minimum and maximum wavelengths (parameters menu)
- Set "Include Arrays?" to "<> Yes" (parameters menu).
- Enter the minimum and maximum offset ranges within which the noise with the above wavelength appears (parameters menu).
- Make sure that the source and receiver arrays are loaded in the submenus of the array window.
- Click on "Stack Response".

Linear noise suppression. This diagnostic is almost identical to the multiple suppression diagnostic. The noise in this case is a number of linear events. The suppression is given by

$$R = \frac{1}{f_{max} - f_{min}} \int_{f_{min}}^{f_{max}} df \left| \frac{1}{N_{LM} N} \sum_{m=1}^{N_{LM}} \sum_{n=1}^{N} e^{2\pi i f t_{nm}} \right|^2 \qquad \text{(EQ 23)}$$

where the NMO-corrected times of the linear events are given by

$$t_{nm} = \sqrt{\left(\frac{x_n}{V_{LM}} + (m-1)\,\Delta t\right)^2 - \frac{x_n^2}{V_P^2}} \qquad \text{(EQ 24)}$$

Usage

- Enter the offset limitations (main menu).
- Enter the linear noise velocity V_{LM} (parameters menu)
- Enter the number of linear events N_{LM} and the time between events Δt (parameters menu)
- Enter the minimum and maximum frequencies (parameters menu)
- Enter the primary velocity (main menu).
- Set "Use Arrays?" to "<> No"(parameters menu).
- Click on "Linear Noise Supp.".

Linear noise suppression with arrays. The wavenumber of the noise is a function of the frequency:

$$k(f) = \frac{2\pi f}{V_{LM}} \qquad \text{(EQ 25)}$$

The linear noise suppression is then

$$R = \frac{1}{f_{\max} - f_{\min}} \int_{f_{\min}}^{f_{\max}} df |R(f)|^2 \qquad \text{(EQ 26)}$$

where

$$R(f) = \frac{1}{N_{LM} N} \sum_{m=1}^{N_{LM}} \sum_{n=1}^{N} e^{2\pi i f t_{nm}} R_R(k(f), \theta_n) R_S(k(f), \theta_n) \qquad \text{(EQ 27)}$$

Usage

- Enter the offset limitations (main menu).
- Enter the linear noise velocity V_{LM} (parameters menu)
- Enter the number of linear events N_{LM} and the time between events Δt (parameters menu)
- Enter the minimum and maximum frequencies (parameters menu)
- Enter the primary velocity (main menu).
- Set "Use Arrays?" to "<> Yes" (parameters menu)
- Make sure the source and receiver arrays are set in the array submenus in the Array window.
- Click on "Linear Noise Supp.".

Stack response plotted versus k. Plots $R(k)$ in every cell currently selected.The scale on the y-axis ranges from 0 dB to -40 dB, and the scale on the x-axis ranges from $\frac{2\pi}{\lambda_{\max}}$ to $\frac{2\pi}{\lambda_{\min}}$.

$$R(k) = \left| \frac{1}{N} \sum_{n=1}^{N_{noise}} e^{ikx_n} \right| \qquad \text{(EQ 28)}$$

Usage

- Enter the offset limitations (main menu).
- Enter the minimum and maximum wavelengths (parameters menu)
- Enter the minimum and maximum offset ranges within which the noise with the above wavelength appears (parameters menu).
- Set "Include Arrays?" to "<> No" (parameters menu).
- Click on "Stack Response vs k".

Note: see the "Stack Response" diagnostic.

Stack response versus k with arrays. The same as above, except that the scale on the y-axis ranges from 0 to -80 dB and

$$R(k) = \left| \frac{1}{N} \sum_{n=1}^{N_{noise}} e^{ikx_n} R_R(k, \theta_n) R_S(k, \theta_n) \right| \qquad \text{(EQ 29)}$$

Usage

- Enter the offset limitations (main menu).
- Enter the minimum and maximum wavelengths (parameters menu)
- Set "Include Arrays?" to "<> Yes"(parameters menu).
- Enter the minimum and maximum offset ranges within which the noise with the above wavelength appears (parameters menu).
- Make sure that the source and receiver arrays are loaded in the submenus of the array window.
- Click on "Stack Response".

Note: see the "Stack Response" diagnostic.

4.7.3 Histograms/Distributions/Azimuth displays

Composite Histogram. Generates composite offset and azimuth histograms for all traces contained within the selected area.

Midpoint plot. Places a small "+" at the midpoint of every trace.

Offset Histogram.

Offset Distribution

Azimuth Histogram.

Azimuth Distribution.

Spider plots.

Offset-Azimuth Scatter Plot.

Each trace in the area selected is represented by a point in this plot. Each point may represent more than one trace within the selected area.

Offset-weighted Average Azimuth

$$\langle\theta\rangle = \frac{\sum_{n=1}^{N} X_n \theta_n}{\sum_{n=1}^{N} X_n} \qquad \text{(EQ 30)}$$

where θ_n is mapped to the range $0 \le \theta \le 90$. (This only makes sense if the survey is shot split-spread). This diagnostic can be useful for detecting rapid variations in the crossline component of the offsets.

4.7.4 Pixmap Storage

Up to ten diagnostics displays, and their associated color histograms, may be saved in memory. After a plot has been generated simply click on one of the "Save Pixmaps" buttons. The plot can be redisplayed by clicking on the "Copy to Display" button. A name for the pixmap may be inserted into the entry field to the right.

Pixmaps use a large amount of memory, so be sure to free pixmaps no longer in use.

4.7.5 Detail window

Cell-by-cell examination.

Usage
- Select "<> Cell Detail".
- Enter the minimum and maximum offsets.
- Click the left button on the cell of interest.

TRK/BIN traverse offset distribution

Usage
- Select "<> Fixed TRK" to display the offsets along a given TRK line, or select "<> Fixed BIN" to display the offsets along a given BIN line.
- Select "<> Display Offsets" to display the magnitude of the offsets, "<>xline offset" to display the only the component of the offsets in the crossline direction, or "<>inline offsets" to display the inline component.
- Select either "<>Color Azimuths" or "<>Black Azimuths".
- Enter the offset range.
- Click the left button on the cell of interest.

DMO point movement

Usage
- Enter the stacking velocity, zero-offset time, dip and azimuth of the reflector. They are found in the parameters menu.
- Select either "<> Fixed TRK" or "<>Fixed BIN".
- Select "<> Disp: xline spray" to show how far the DMO points will move in the crossline direction, or "<> Disp: inline spray" to show how far they will move in the inline direction.
- Enter the minimum and maximum expected movements. The minimum will usually be negative, and corresponds to the negative x or y coordinate direction.
- Click the left button on the track or bin line of interest.

4.7.6 DMO points binning

It is possible to create DMO points hit structures (representing the surface locations of the equivalent zero-offset traces after DMO) directly from the CMP hit structures. The routine

that does this assumes that a unit cell has been binned, and that the unit cell is "surrounded" by other unit cells so that edge effects do not occur. For example, if a DMO point is moved off the right edge of the area binned, then the routine assumes that a corresponding trace moves into the region binned from the left edge.

Usage

• Enter the stacking velocity, zero-offset time, dip and azimuth of the reflector. They are found in the parameters menu.
• At the top of the main menu click on "CMP -> DMO".
• Select either "<>Use CMP" or "<>Use DMO".

Both hit structures are retained in memory. Note that certain diagnostics are disabled when "<> Use DMO" has been selected.

4.8 Array Button

This module allows the interactive design of source and receiver arrays. Once the arrays have been designed they may be incorporated into the coherent noise suppression diagnostics.

4.8.1 Array specification methods

There are two methods of specifying the x-y coordinates of the array elements

•Method 1: direct entry into the source and receiver menus

This method of entry is useful if the array elements do not lie on a regular grid.

Usage

• Open the source (or receiver) menu and directly enter the x-coordinate, y-coordinate, and weight of each element.
• Set the "Use?" option to yes for each element entered.
• Close the source/receiver menu and open the plot menu. Near the bottom of the plot menu set "Auto Scale" to "<> Yes".
• Click "Plot Src/Rcvr Array".

•Method 2: Entry via the mouse

This method works best when the array elements lie on a regular grid.

Usage

• Open the plot menu.
• Set "Auto Scale" to "<> NO".
• Enter the grid size (usually the spacing between elements).
• Specify the greatest extent of the array (relative to the center of the array).
• Click "Plot Src(Rcvr) Array)".
• If every array element lies on a grid intersection then set "Snap to Grid" to "<>YES".
• Move the cursor to the array plot and click on the middle mouse button.
• The left button can be used to delete array elements

The array elements are added to the source or receiver menu as they added to the plot.

4.8.2 Automatic array scaling

The souce and receiver arrays may be automatically rescaled.

Usage

• Open the source (or receiver) menu.
• Enter the appropriate scale factors.
• Click on "Apply Factors"

The array coordinates are modified as follows:

$$x_{new} = x_{old} x_{factor} \text{ and } y_{new} = y_{old} y_{factor} \qquad \text{(EQ 31)}$$

4.8.3 Saving the arrays

Usage

- Open the source (or receiver) menu
- Enter a file name
- Click on "Store Array"

4.8.4 Array response calculations

4.8.5 Surface noise attenuation

The noise is assumed to be plane wave in nature, and so independent of the separation between the source and receiver arrays.

The source and receiver array responses are calculated as follows. Let (x_n, y_n) be the set of array elements (for either array) and define

$$d_n = \sqrt{x_n^2 + y_n^2} \text{ and } \phi_n = \operatorname{atan}\left(\frac{y_n}{x_n}\right) \text{ where } 0 \le \phi_n \le 2\pi \qquad \text{(EQ 32)}$$

The noise can be described using either (k_x, k_y) or (λ, θ) where

$$\lambda = \frac{2\pi}{\sqrt{k_x^2 + k_y^2}} \text{ and } \theta = \operatorname{atan}\left(\frac{k_y}{k_x}\right) \qquad \text{(EQ 33)}$$

The projection of an array element along the noise direction is given by

$$p_n = d_n \cos(\phi_n - \theta) \qquad \text{(EQ 34)}$$

The response of an array to signal with a given wavelength and azimuth is

$$R(\lambda, \theta) = \left| \frac{\sum_{n=1}^{N} w_n e^{2\pi i \frac{p_n}{\lambda}}}{\sum_{n=1}^{N} w_n} \right| \qquad \text{(EQ 35)}$$

where w_n is the weight of the nth array element.

Usage

- Select the type of display, either "<>Kx vs. Ky" or "<>Wave. vs theta".
- Select a wavelength range.
- Select the display parameters: the number of intervals, number of colors, and dB per color.
- Click on either "Source Response", "Receiver Response", or "Src/Rec Response"

Note: The kx and ky displayed on the screen have been divided by 2π to represent $\frac{1}{\lambda}$.

Detailed information regarding any point on the display may be obtained by positioning the cursor at the desired coordinate.

4.8.6 Surface noise attenuation with a "pinch of salt"

Uncertainties in the exact locations of the array elements may be incorporated in the above diagnostics[6]. The errors in the array positions are asssumed to be independent Gaussian random variables with zero mean and standard deviation (σ_x, σ_y). The expected response is

$$\langle |R(k_x, k_y)|^2 \rangle = |R(k_x, k_y)|^2 \beta + (1-\beta) \sum_{n=1}^{N} w_n^2 \qquad \text{(EQ 36)}$$

where

$$\beta = \exp(-\sigma_x^2 k_x^2 - \sigma_y^2 k_y^2) \qquad \text{(EQ 37)}$$

This response is then normalized by the sum of the weights squared.

Usage

- Select the type of display, either "<>Kx vs. Ky" or "<>Wave. vs theta".
- Select a wavelength range.
- Select the display parameters: the number of intervals, number of colors, and dB per color.
- Set "Include Uncertainties" to "Yes".
- Enter the standard deviations for the source and/or receiver array elements.
- Click on either "Source Response", "Receiver Response", or "Src/Rec Response"

4.8.7 Desired signal attenuation

The source and receiver arrays should not attenuate the desired signals. The signal is described by NMO:

$$t(x) = \sqrt{t_0^2 + \frac{x^2}{V^2}} \qquad \text{(EQ 38)}$$

The apparent velocity of the signal at the surface is

$$V_{\text{app}}(x) = \frac{dx}{dt} = V\sqrt{1 + \frac{V^2 t_0^2}{x^2}} \qquad \text{(EQ 39)}$$

The apparent wavenumber of the signal is a function of the offset and the frequency:

$$k(x, f) = \frac{2\pi f}{V_{\text{app}}(x)} \qquad \text{(EQ 40)}$$

The suppression as a function of offset and frequency is

$$R(x, f) = R_S(x, f) R_R(x, f) \qquad \text{(EQ 41)}$$

where

$$R_S(x,f) = \left| \frac{\sum_{n=1}^{N} w_n e^{ik(x,f)p_n}}{\sum_{n=1}^{N} w_n} \right| \qquad \text{(EQ 42)}$$

with a similiar equation for the receiver array.

Usage

- Select "<>Offset vs. freq".
- Enter the offset and frequency ranges.
- Enter the NMO velocity and zero-offset time.
- Select a source array-receiver array azimuth.
- Select the display parameters: the number of intervals, the number of colors, and the dB per color. The dB per color should be fairly small (usually about 0.5).
- Click on "Src/Rec Response".

The suppression should be checked for a range of azimuths.

References

1. Holley, T. K. (1990), 3DSIGN: The Shell 3D Survey Design Workstation,Technical Progress Report BRC 77-90, Shell Development Bellaire Research Center, Houston, November.
2. Holley, T. K., Ensign, P. W. and Parrish, J. F. (1990), Multiple Suppression by Stacking in Land 3-D Surveys, Part I: Modeling with the Interactive Survey Design Workstation, Exploration Bimonthly Research Summary, Shell Development Bellaire Research Center, Houston, January.
3. Ensign, P. W., Holley, T. K., and Parrish, J. F. (1990), Multiple Suppression by Stacking in Land 3-D Surveys, Part II: Case History of Omar, Syria, 1988 and 1989, Exploration Bimonthly Research Summary, Shell Development Bellaire Research Center, Houston January.
4. Parrish, J. F. (1989), Multiple Suppression in Land 3D Surveys: Summarizing the Unit Cell of a Survey, Item 49-13238, Exploration Bimonthly Research Summary, Shell Development Bellaire Research Center, Houston, January, p. 51.
5. Parrish, J. F. (1989), Techniques for analyzing the Effect of Acquisition Geometry for Land 3-D Surveys: Part I: Factoring and Scaling the Stacked Response, Paper 47, 1989 Geophysical Conference, Shell Development Bellaire Research Center, Houston, January.
6. Newman, P. and Mahoney, J.T. (1973) Patterns - with a pinch of salt. Geophysical Prospecting, Volume 21, pages 197-219.